

Linear classification

Classification: Classification is a supervised machine learning technique used to predict labels or categories from input data. It assigns each data point to a predefined class based on learned patterns.

For example: A classification model might be trained on dataset of images labeled as either dogs or cats and it can be used to predict the class of new and unseen images as dogs or cats based on their features such as colour, texture or shape.

Types of Classification

Classification in machine learning involves sorting data into categories based on their features or characteristics. The type of classification problem depends on how many classes exist and how the categories are structured.

1. Binary Classification

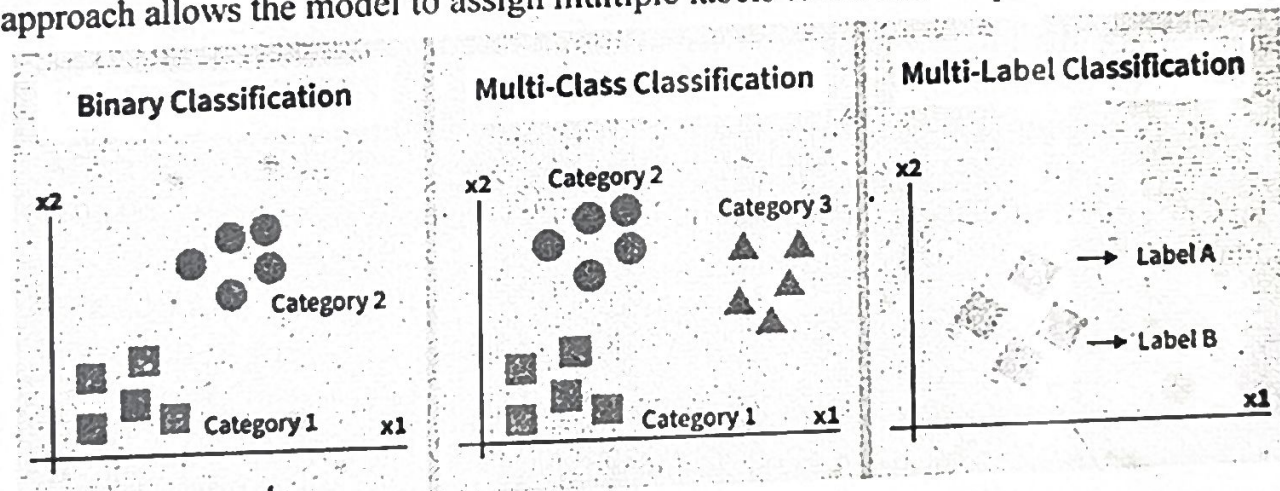
Binary classification is the simplest type of classification where data is divided into two possible categories. The model analyzes input features and decides which of the two classes the data belongs to.

2. Multiclass Classification

Multiclass classification is used when data needs to be divided into more than two categories. The model analyzes the input features and selects the class that best matches the data.

3. Multi Label Classification

Multi label classification allows a single piece of data to belong to multiple categories at the same time. Unlike multiclass classification, where each data point is assigned only one class, this approach allows the model to assign multiple labels to the same input.



Working

Classification works by training a model on labeled data so it can learn patterns and predict the correct class for new inputs. The main steps involved are:

1. **Data Collection:** Start with a dataset where each data point has a correct label.
2. **Feature Extraction:** Identify important features such as color, shape or texture that help distinguish classes.
3. **Model Training:** The algorithm learns patterns that connect features to the correct class.
4. **Model Evaluation:** The trained model is tested on unseen data to measure its accuracy.
5. **Prediction:** The model predicts the class of new data based on learned patterns.

6. **Model Improvement:** If performance is not satisfactory, the model or its parameters are adjusted and retrained.

Classification Algorithms

To implement a classification model, it is important to understand the algorithms used for classification. One of the most commonly used algorithms is Logistic Regression. Classification algorithms can be grouped into different categories, such as:

1. Linear Classifiers

Linear classifier models create a linear decision boundary between classes. They are simple and computationally efficient. A linear classifier makes classification decisions based on a linear combination of input features. It draws a straight line, plane, or hyperplane to separate data points into different classes. Some of the linear classification models are as follows:

- Logistic Regression
- Support Vector Machines
- Single-layer Perceptron

How It Works

- **Feature Vector (x):** The input data represented as numbers (like size, weight, or pixel values).
- **Weights (w) and Bias (b):** Parameters learned by the model during training that determine the orientation and position of the boundary.
- **Linear Score (z):** Calculated as the dot product $(z = w^T x + b)$.
- **Decision Boundary:** The threshold where the score equals zero, dividing one class from another:

Strengths and Limitations

- **Fast and Efficient:** Requires low computing power to train and make predictions, making it great for large datasets with many variables.
- **Easy to Interpret:** Inspecting the weight values shows which features matter most for a decision.
- **Cannot Handle Complex Patterns:** Fails if data is heavily overlapped or requires a curved, non-linear boundary (like an XOR logic problem).

2. Non linear Classifiers

Non linear models create non linear decision boundaries to separate classes. They can capture more complex relationships between input features and the target variable. Some common non linear classification models include:

- K-Nearest Neighbours
- Kernel SVM
- Decision Tree Classification
- Random Forests
- AdaBoost
- Bagging Classifier
- Voting Classifier
- Extra Trees Classifier

Logistic Regression:

Logistic regression is a supervised machine learning algorithm mainly used for classification tasks where the goal is to predict the probability that an instance of belonging to a given class. It is used for classification algorithms its name is logistic regression. It's referred to as regression because it takes the output of the linear regression function as input and uses a sigmoid function to estimate the probability for the given class. The difference between linear regression and logistic regression is that linear regression output is the continuous value that can be anything while logistic regression predicts the probability that an instance belongs to a given class or not.

It is used for predicting the categorical dependent variable using a given set of independent variables:

- Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value.
- It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.
- Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas Logistic regression is used for solving the classification problems.
- In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).
- The curve from the logistic function indicates the likelihood of something such as whether the cells are cancerous or not, a mouse is obese or not based on its weight, etc.
- Logistic Regression is a significant machine learning algorithm because it has the ability to provide probabilities and classify new data using continuous and discrete datasets.
- Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification.

Logistic Function (Sigmoid Function):

- The sigmoid function is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1. The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form.
- The S-form curve is called the Sigmoid function or the logistic function.

- In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.

Type of Logistic Regression:

On the basis of the categories, Logistic Regression can be classified into three types:

- Binomial:** In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
- Multinomial:** In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as "cat", "dogs", or "sheep"
- Ordinal:** In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as "low", "Medium", or "High".

Sr.No	Linear Regression	Logistic Regression
1	Linear regression is used to predict the <u>continuous dependent variable</u> using a given set of independent variables.	Logistic regression is used to predict the <u>categorical dependent variable</u> using a given set of independent variables.
2	Linear regression is used for <u>solving Regression problem.</u>	It is used for <u>solving classification problems.</u>
3	In this we <u>predict the value of continuous variables</u>	In this we <u>predict values of categorical variables</u>
4	In this we find <u>best fit line.</u>	In this we find <u>S-Curve</u> .
5	Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for Estimation of accuracy.
6	The output must be <u>continuous value</u> , such as price, age, etc.	Output is must be <u>categorical value</u> such as 0 or 1, Yes or no, etc.
7	It required <u>linear relationship between dependent and independent variables.</u>	It <u>not required linear relationship.</u>

Sr.No	Linear Regression	Logistic Regression
8	There may be collinearity between the independent variables.	There should not be collinearity between independent variable.

How does Logistic Regression work?

The logistic regression model transforms the linear regression function continuous value output into categorical value output using a sigmoid function, which maps any real-valued set of independent variables input into a value between 0 and 1. This function is known as the logistic function.

Let the independent input features be

$$X = \begin{bmatrix} x_{11} & \dots & x_{1m} \\ x_{21} & \dots & x_{2m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nm} \end{bmatrix}$$

and the dependent variable is Y having only binary value i.e. 0 or 1.

$$Y = \begin{cases} 0 & \text{if Class 1} \\ 1 & \text{if Class 2} \end{cases}$$

then apply the multi-linear function to the input variables X

$$z = (\sum_{i=1}^n w_i x_i) + b$$

Here x_i is the i th observation of X, $w_i = [w_1, w_2, w_3, \dots, w_m]$ is the weights or Coefficient, and b is the bias term also known as intercept. simply this can be represented as the dot product of weight and bias.

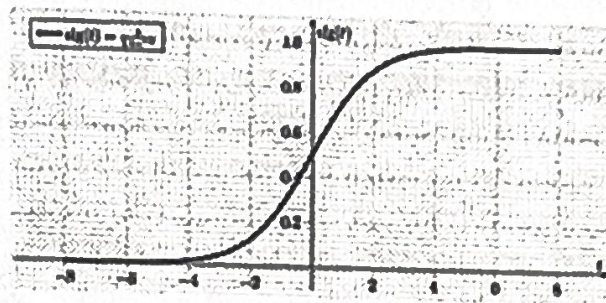
$$z = w \cdot X + b$$

whatever we discussed above is the linear regression.

Sigmoid Function

Now we use the sigmoid function where the input will be z and we find the probability between 0 and 1. i.e predicted y .

$$\sigma(z) = \frac{1}{1+e^{-z}}$$



Sigmoid function

As shown above, the figure sigmoid function converts the continuous variable data into the probability i.e. between 0 and 1.

- $\sigma(z)$ tends towards 1 as $z \rightarrow \infty$
- $\sigma(z)$ tends towards 0 as $z \rightarrow -\infty$
- $\sigma(z)$ is always bounded between 0 and 1

Cost function:
↳ LR generally uses log loss rather than the ordinary least-square error

$$J : - \sum [y \log(p) + (1-y) \log(1-p)]$$

$$J(\beta) = -\frac{1}{m} \sum_{i=1}^m [y_i \log p_i + (-y_i) \log(1-p_i)]$$

$h(x_i)$

Ex

$$x_1 = \text{temp.}$$

$$x_2 = \text{humidity}$$

$$z = -10 + 0.25x_1 + 0.08x_2$$

$$x_1 = 25$$

$$x_2 = 80$$

$$\begin{aligned} \therefore z &= -10 + (0.25)(25) + (0.08)(80) \\ &= -10 + 6.25 + 6.4 \\ &= 2.64 \end{aligned}$$

$$P(y=1) = \frac{1}{1 + e^{-2.64}} = 0.937$$

Predicted probability of thunderstorm 93.4%.
which is greater than 0.5, model predict thunderstorm.

Gradient descent minimize the cost

↳ After calculating the cost, LR adjust its parameter using optimization algo such as Gradient descent.

$$\theta_j = \theta_j - \alpha \frac{\partial J\theta}{\partial \theta_j}$$

where θ_j = model parameter

α = learning rate

$J(\theta)$ = cost function

$\frac{\partial J}{\partial \theta_j}$ = gradient of cost

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i) x_{ij}$$

$$\theta_j \leftarrow \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i) x_{ij}$$

Input data $\rightarrow$ linear equation $\rightarrow$ sigmoid $\rightarrow$ probability
 $\rightarrow$ log loss $\rightarrow$ gradient $\rightarrow$ update